Strange Way to Express Integers

Time Limit/Memory Limit: 1000ms/128M

Total Submissions/Accepted: 28377/9074

Language: C++98 G++/MVSC

Description

Elina is reading a book written by Rujia Liu, which introduces a strange way to express non-negative integers. The way is described as following:

Choose k different positive integers $a_1, a_2, \ldots, a_k$. For some non-negative m, divide it by every $a_i (1 \leq i \leq k)$ to find the remainder r_i. If $a_1, a_2, \ldots, a_k$ are properly chosen, m can be determined, then the pairs (a_i, r_i) can be used to express m.

“It is easy to calculate the pairs from m, ” said Elina. “But how can I find m from the pairs?”

Since Elina is new to programming, this problem is too difficult for her. Can you help her?

Input

The input contains multiple test cases. Each test cases consists of some lines.

Line 1: Contains the integer k.

Lines 2 ~ $k + 1$: Each contains a pair of integers $a_i, r_i (1 \leq i \leq k)$.

Output

Output the non-negative integer m on a separate line for each test case. If there are multiple possible values, output the smallest one. If there are no possible values, output -1.

Sample

Input

```
2
8 7
11 9
```

Output

```
31
```

Hint

All integers in the input and the output are non-negative and can be represented by $64 - bit$ integral types.

Source

POJ Monthly--2006.07.30, Static.